

OS-PRACTICAL

Practical-8

1. Develop a program using malloc(), calloc(), realloc(), and free(). Monitor memory allocation behavior and identify memory leaks using Valgrind.

Develop a program that demonstrates Copy-on-Write behavior after a fork(). Analyze memory usage before and after modifying data in the child process.

```
ananya@DESKTOP-42I5F93: ~$ mkdir practical8
ananya@DESKTOP-42I5F93: ~$ cd practical8
ananya@DESKTOP-42I5F93: ~/practical8$ nano memory_alloc.c
ananya@DESKTOP-42I5F93: ~/practical8$ gcc memory_alloc.c -o memory_alloc
ananya@DESKTOP-42I5F93: ~/practical8$ ./memory_alloc
malloc completed
calloc completed
realloc completed
free completed
ananya@DESKTOP-42I5F93: ~/practical8$ valgrind --version
valgrind-3.26.0
ananya@DESKTOP-42I5F93: ~/practical8$ valgrind --leak-check=full ./memory_alloc
==670== Memcheck, a memory error detector
==670== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==670== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==670== Command: ./memory_alloc
==670==
malloc completed
calloc completed
realloc completed
free completed
==670==
==670== HEAP SUMMARY:
==670==   in use at exit: 20 bytes in 1 blocks
==670==   total heap usage: 4 allocs, 3 frees, 1,104 bytes allocated
==670==
==670== 20 bytes in 1 blocks are definitely lost in loss record 1 of 1
==670==    at 0x4850858: malloc (vg_replace_malloc.c:447)
==670==    by 0x40011DE: main (in /home/ananya/practical8/memory_alloc)
==670==
==670== LEAK SUMMARY:
==670==    definitely lost: 20 bytes in 1 blocks
==670==    indirectly lost: 0 bytes in 0 blocks
==670==    possibly lost: 0 bytes in 0 blocks
==670==    still reachable: 0 bytes in 0 blocks
==670==    suppressed: 0 bytes in 0 blocks
==670==
==670== For lists of detected and suppressed errors, rerun with: -s
==670== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
ananya@DESKTOP-42I5F93: ~/practical8$ nano memory_alloc.c
```

```
ananya@DESKTOP-42I5F93:~/practical8$ nano memory_alloc.c
ananya@DESKTOP-42I5F93:~/practical8$ gcc memory_alloc.c -o memory_alloc
ananya@DESKTOP-42I5F93:~/practical8$ ./memory_alloc
malloc completed
calloc completed
realloc completed
free completed
ananya@DESKTOP-42I5F93:~/practical8$ valgrind --leak-check=full ./memory_alloc
==733== Memcheck, a memory error detector
==733== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==733== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==733== Command: ./memory_alloc
==733==
malloc completed
calloc completed
realloc completed
free completed
==733==
==733== HEAP SUMMARY:
==733==     in use at exit: 0 bytes in 0 blocks
==733==   total heap usage: 4 allocs, 4 frees, 1,104 bytes allocated
==733==
==733== All heap blocks were freed -- no leaks are possible
==733==
==733== For lists of detected and suppressed errors, rerun with: -s
==733== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
ananya@DESKTOP-42I5F93:~/practical8$ nano cow.c
ananya@DESKTOP-42I5F93:~/practical8$ gcc cow.c -o cow
ananya@DESKTOP-42I5F93:~/practical8$ ./cow
Before fork: data[0] = 10
Child before modification: data[0] = 10
Child after modification: data[0] = 20
Parent after child modification: data[0] = 10
```